$$\frac{\partial F^{\alpha\beta}}{\partial x^\beta} = -\mu_0 j^\alpha = 0$$ in the absence of any charges (that is, in vacuum).

$$\Rightarrow \frac{\partial}{\partial x^\beta}\left(\frac{\partial A^\beta}{\partial x_\alpha} - \frac{\partial A^\alpha}{\partial x_\beta}\right) = 0$$

$$\Rightarrow (1)\ \frac{\partial^2 A^\beta}{\partial x_\alpha \partial x^\beta} - \frac{\partial^2 A^\alpha}{\partial x_\beta \partial x^\beta} = 0$$

Note: β is repeated $\rightarrow$ summed over. $\alpha = 0, 1, 2, 3 \Rightarrow$ 4 equations in total.

But there is gauge freedom: can add $\frac{\partial g}{\partial x_\alpha}$ to form

$$\tilde{A}^\alpha = A^\alpha + \frac{\partial g}{\partial x_\alpha}$$. This does not change $\vec{E}, \vec{B}$

and $$\tilde{F}^{\alpha\beta} = \frac{\partial \tilde{A}^\beta}{\partial x_\alpha} - \frac{\partial \tilde{A}^\alpha}{\partial x_\beta} = \frac{\partial}{\partial x_\alpha}\left(A^\beta + \frac{\partial g}{\partial x_\beta}\right) -$$

$$- \frac{\partial}{\partial x_\beta}\left(A^\alpha + \frac{\partial g}{\partial x_\alpha}\right) = \underbrace{\frac{\partial A^\beta}{\partial x_\alpha} - \frac{\partial A^\alpha}{\partial x_\beta}}_{F^{\alpha\beta}} + \frac{\partial^2 g}{\partial x_\alpha \partial x_\beta} - \frac{\partial^2 g}{\partial x_\beta \partial x_\alpha}$$

$$\Rightarrow \tilde{F}^{\alpha\beta} = F^{\alpha\beta}.$$

Thus, $F^{\alpha\beta}$ does not change (is invariant under the gauge transformation)

therefore $\frac{\partial}{\partial x^\beta} F^{\alpha\beta} = 0$ also does not change.

We can use such a gauge freedom to simplify Eq.(1).

Namely, we can always choose A^α to satisfy <u>Lorentz gauge</u>

$$\frac{\partial A^\alpha}{\partial x^\alpha} = 0.$$

If so, then (1) turns into $\dfrac{\partial^2 A^\alpha}{\partial x_\beta \partial x^\beta} = 0$

$$\Rightarrow \frac{1}{c^2}\frac{\partial^2 A^\alpha}{\partial t^2} - \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) A^\alpha = 0$$

or, $\dfrac{1}{c^2}\dfrac{\partial^2 A^\alpha}{\partial t^2} - \nabla^2 A^\alpha = 0$

or, $\Box = \nabla^2 - \dfrac{1}{c^2}\dfrac{\partial^2}{\partial t^2}$ (D'Alembertian)

$\Box A^\alpha = 0$. This holds for every α.

<u>Clarification</u>: how can we make sure that $\dfrac{\partial A^\alpha}{\partial x^\alpha} = 0$ holds?

Suppose we have A^α such that $\dfrac{\partial A^\alpha}{\partial x^\alpha} \neq 0$.

Then we introduce $\tilde{A}^\alpha = A^\alpha + \dfrac{\partial g}{\partial x_\alpha}$ and require that $\dfrac{\partial \tilde{A}^\alpha}{\partial x^\alpha} = 0$

$\Rightarrow$ but $\dfrac{\partial \tilde{A}^\alpha}{\partial x^\alpha} = \dfrac{\partial A^\alpha}{\partial x^\alpha} + \dfrac{\partial^2 g}{\partial x^\alpha \partial x_\alpha}$

$\Rightarrow$ we obtain differential equation on function

$$g: \quad \frac{\partial^2 g}{\partial x_\alpha \partial x^\alpha} = -\frac{\partial A^\alpha}{\partial x^\alpha}.$$

Explicitly, $\underbrace{\nabla^2 g - \dfrac{1}{c^2}\dfrac{\partial^2 g}{\partial t^2} = \dfrac{\partial A^\alpha}{\partial x^\alpha}}$.

$g(\vec{r}, t)$ must satisfy

Then $\tilde{A}^\alpha$ will satisfy $\dfrac{\partial \tilde{A}^\alpha}{\partial x^\alpha} = 0$ and $\Box \tilde{A}^\alpha = 0$.

Consider now a <u>plane wave</u> solution of $\Box A^\alpha = 0$.

let $A^\alpha(\vec{r},t) = A_0^\alpha f(x-vt)$

$\hookrightarrow$ 4-vector amplitude (constant).

$$\Rightarrow \quad \nabla^2 A^\alpha(\vec{r},t) = A_0^\alpha \frac{\partial^2}{\partial x^2} f(x-vt) = A_0^\alpha f''(x-vt)$$

$$\frac{\partial^2 A^\alpha(\vec{r},t)}{\partial t^2} = A_0^\alpha \frac{\partial^2}{\partial t^2} f(x-vt) = A_0^\alpha f''(x-vt) \cdot \left(\frac{\partial}{\partial t}(x-vt)\right)^2.$$

$$\Rightarrow \quad \left(\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right) A^\alpha(\vec{r},t) = 0 \quad \text{gives}$$

$$A_0^\alpha \left(1 - \frac{v^2}{c^2}\right) f''(x-vt) = 0.$$

This is satisfied by any non-zero function f provided that $v^2 = c^2 \Rightarrow v = \pm c$.

$\Rightarrow$ for $v = +c$ $f(x-vt) = f(x-ct)$ describes right-moving wave.

$v = -c$ $f(x-vt) = f(x+ct)$ — left-moving wave.

All waves propagate with the speed of light c.

Last lecture: back to wave equation

$$\frac{1}{c^2}\frac{\partial^2 A^\alpha}{\partial t^2} - \nabla^2 A^\alpha = 0 \quad (1)$$

$$\vec{E} = -\vec{\nabla}\varphi - \frac{\partial \vec{A}}{\partial t} \quad , \vec{B} = \vec{\nabla}\times\vec{A}$$

Since (1) is linear, $\vec{B}$ and $\vec{E}$ satisfy the same eq.

$$\Rightarrow \vec{E}(\vec{r},t) = \vec{E}_0\, e^{i\vec{k}\cdot\vec{r} - i\omega t}$$

$$\vec{B}(\vec{r},t) = \vec{B}_0\, e^{i\vec{k}\cdot\vec{r} - i\omega t}$$

$$\left.\begin{aligned}\vec{\nabla}\times\vec{E} &= -\frac{\partial\vec{B}}{\partial t}\\ \vec{\nabla}\times\vec{B} &= \frac{1}{c^2}\frac{\partial\vec{E}}{\partial t}\end{aligned}\right\} \quad \begin{cases} i\vec{k}\times\vec{E} = i\omega\vec{B}\\ i\vec{k}\times\vec{B} = \frac{-i\omega}{c^2}\vec{E}\end{cases}$$

$$\left.\begin{aligned}\vec{\nabla}\cdot\vec{E} &= 0\\ \vec{\nabla}\cdot\vec{B} &= 0\end{aligned}\right\} \Rightarrow \begin{aligned} & i\vec{k}\cdot\vec{E}_0\, e^{i\vec{k}\vec{r} - i\omega t} = i\vec{k}\cdot\vec{E}(\vec{r},t) = 0\\ & i\vec{k}\cdot\vec{B} = 0\end{aligned}$$

$\vec{E}, \vec{B}$ are transverse to $\vec{k}$.

And $\vec{E}\cdot\vec{B} = 0$.

$\Rightarrow$

$$i\vec{k}\times\vec{E} = i\left(-\frac{i\omega}{c^2}\right)^{-1} i\vec{k}\times(\vec{k}\times\vec{B}) = -\frac{c^2}{\omega} i\left[\vec{k}\underbrace{(\vec{k}\cdot\vec{B})}_{=0} - \vec{B}k^2\right]$$

$$= i\frac{c^2 k^2}{\omega}\vec{B} = i\omega\vec{B} \Rightarrow \omega = ck.$$

Constant phase surface $\phi = \vec{k}\cdot\vec{r} - \omega t = \text{const}$

$$\Rightarrow \vec{r}\cdot\hat{k} = r_{\parallel} = \frac{\omega}{k}t + \text{const}$$

It moves with $\frac{\omega}{k} = c$ along $\vec{k}$ direction.

Suppose now that EM emitter is moving with $\vec{V}$.
In the rest frame K' of the emitter $\omega' = \omega_0 = ck'$.

In the lab frame K of the observer

$$E'_{\parallel} = E_{\parallel} \quad , B'_{\parallel} = B_{\parallel}$$

$$\vec{E}'_{\perp} = \gamma\left(\vec{E}_{\perp} + \vec{v} \times \vec{B}\right)$$

$$\vec{B}'_{\perp} = \gamma\left(\vec{B}_{\perp} - \frac{1}{c^2}\vec{v} \times \vec{E}\right)$$

Since $E, B \sim e^{i\vec{k}\cdot\vec{r} - i\omega t}$ and Lorentz transformation connecting $\vec{E}, \vec{B}$ with $\vec{E}', \vec{B}'$ is linear it means that

$$\phi' = \phi \quad \text{the phase is invariant.}$$

(Indeed, $\Delta\phi/2\pi$ counts maxima/minima between any two space-time points — and this number does not depend on the frame of reference)

But $\phi = -\omega t + \vec{k}\cdot\vec{r} \Rightarrow -\phi = \omega t - \vec{k}\cdot\vec{r} = \left(\frac{\omega}{c}, \vec{k}\right)\cdot\left(ct, -\vec{r}\right)$

$$= k^i x_i$$

4-vector $k^i = \left(\frac{\omega}{c}, \vec{k}\right)$, $x^i = (ct, \vec{r})$, $x_i = (ct, -\vec{r})$

<u>Lorentz transformation</u> (let now $\vec{v} = v x$)

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{\omega'}{c} \\ k'_x \end{pmatrix} = \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} \frac{\omega}{c} \\ k_x \end{pmatrix}$$

$$\frac{\omega'}{c} = \gamma\left(\frac{\omega}{c} - \beta k_x\right) = \frac{1}{c}\,\frac{\omega - V k_x}{\sqrt{1 - V^2/c^2}}$$

$$k_x = \frac{\omega}{c}\cos\theta$$

$$\Rightarrow \quad \omega_0 = \frac{\left(\omega - \frac{\omega}{c}\cos\theta\right)}{\sqrt{1 - V^2/c^2}}$$

$$\omega = \omega_0\,\frac{\sqrt{1 - V^2/c^2}}{1 - \frac{V}{c}\cos\theta}$$

Doppler effect: let $\frac{V}{c} \ll 1 \Rightarrow \omega \approx \omega_0\left(1 + \frac{V}{c}\cos\theta\right)$

$\Rightarrow$ frequency of the wave ω increases when EM emitter approaches the observer $V_{\parallel} = V\cos\theta > 0$

$\Rightarrow$ ω decreases when EM source is moving away, $V_{\parallel} = V\cos\theta < 0$.

If $\theta = \pi/2$ $(\cos\theta = 0)$ then $\omega = \omega_0\left(1 - \frac{V^2}{2c^2}\right)$

the frequency gets smaller (transverse Doppler).